

Week 10
Mobile robot kinematics

ENN581 Robot Motion, Control and Planning

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School of Electrical Engineering and Robotics

ACKNOWLEDGEMENT OF TRADITIONAL OWNERS

QUT acknowledges the Turrbal and Yugara, as the First Nations owners of the lands where QUT now stands. We pay respect to their Elders, lores, customs and creation spirits. We recognise that these lands have always been places of teaching, research and learning.

QUT acknowledges the important role Aboriginal and Torres Strait Islander people play within the QUT community.

Exam Prep Tips: Week 10

Refine your 'cheat' sheet (or start one)

- One page that summarises the main ideas and formulas for the exam.
- Helpful, even if not permitted in the exam.
- Watch **this video** (9 min.) to learn how to make a great cheat sheet.



Make a study plan

- Work back from your exam date.
- Plan to revise each topic a few times to improve information retention.
- Focus on the topics you find difficult.
- Watch **this video** (7 min.) for effective planning approaches.



See the full list of tips in the **Guide to Succeeding in Engineering Exams**

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What did we learn so far?

Forward Kinematics: Given the positions of the joints, where is the end effector?

$$^{0}\mathbf{T}_{n} \text{ or } \boldsymbol{\xi}_{n} = \mathcal{K}(\mathbf{q}),\ \mathbf{q} = \{\mathrm{q}_{i}, i \in [1, \cdots, n]\}$$

Inverse Kinematics: Given the tool pose, what are the joint positions to get there?

$$\mathbf{q} = \mathcal{K}^{-1}(\boldsymbol{\xi}_{n})$$

Velocity Kinematics: How fast is the robot moving (FK) or should it go (IK)?

$$^{0}\boldsymbol{\nu} = {}^{0}\mathbf{J}(\mathbf{q})\dot{\mathbf{q}}$$

Trajectory generation: Along what path should a manipulator move?

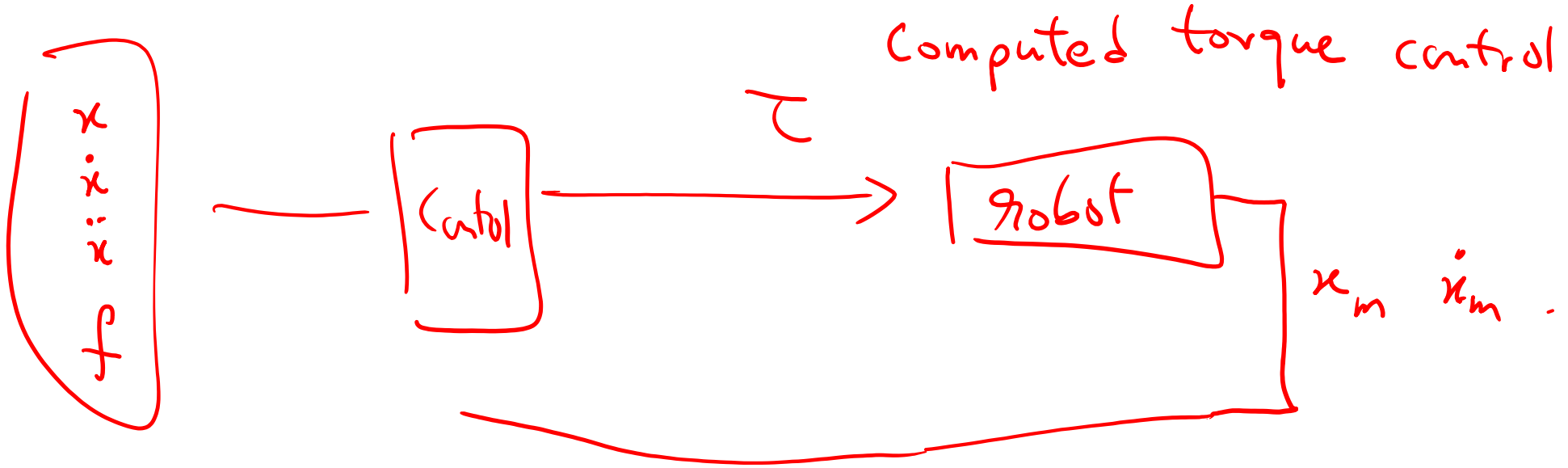
$$\mathbf{q}(t) \text{ and } \boldsymbol{\xi}_{\boldsymbol{n}}(t)$$

Dynamics (inverse): What are the torques at the joint, given a trajectory?

$$\mathbf{Q} = \mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) + \boldsymbol{g}(\mathbf{q})$$

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In the previous two lectures, we looked at low level controllers for both position and force



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But how do we use this on a robot?

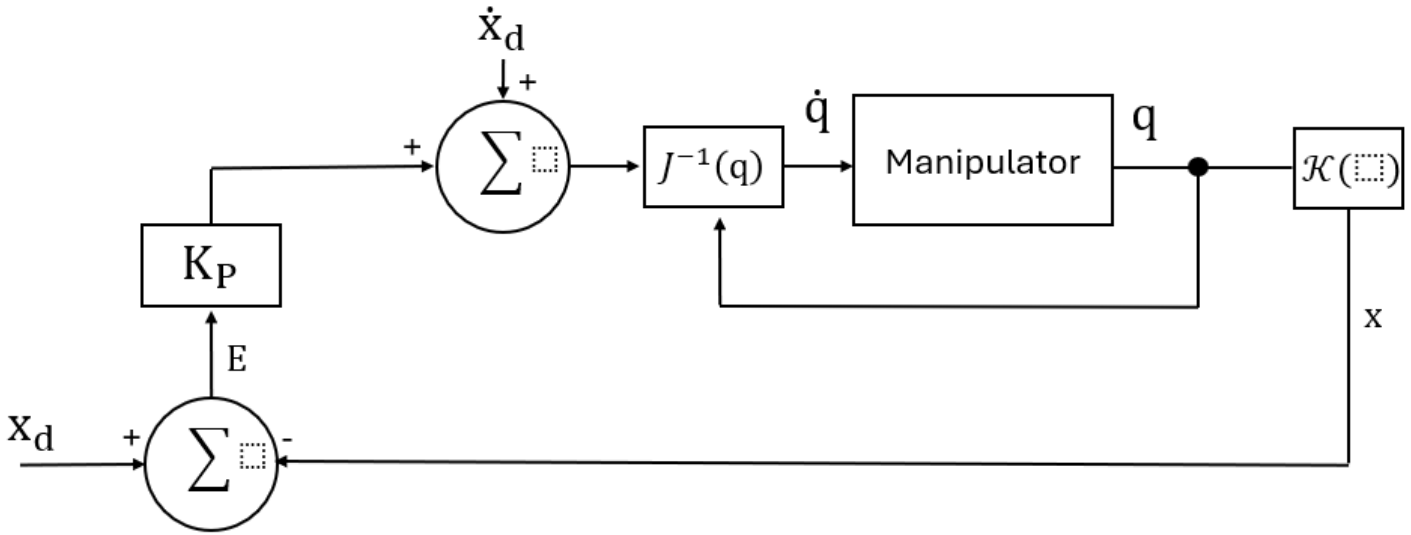


There is no command to set_joint_torque

See the python API

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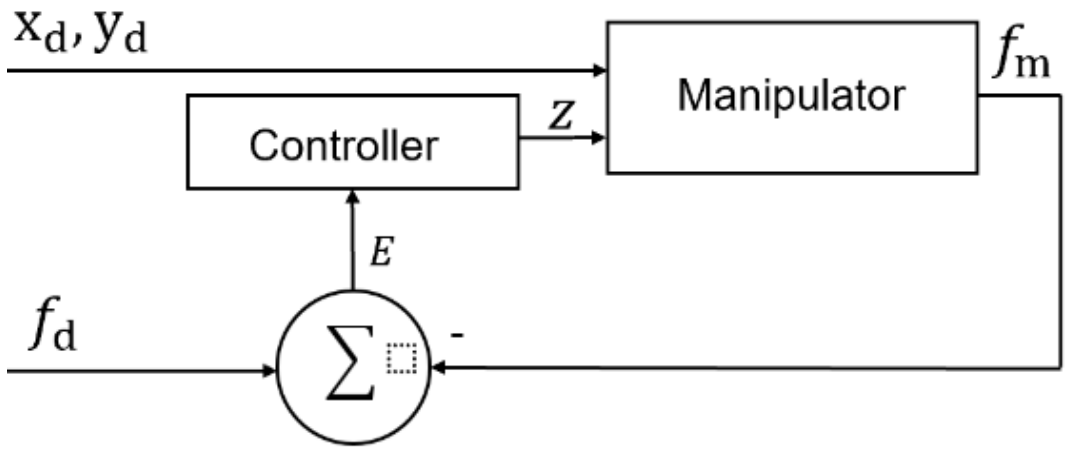
Resolved rate motion control



$$\dot{\boldsymbol{q}} = J^{-1}(\boldsymbol{q})[\dot{\boldsymbol{x}}_{\boldsymbol{d}} + K_P(\boldsymbol{x}_{\boldsymbol{d}} - \boldsymbol{x})]$$

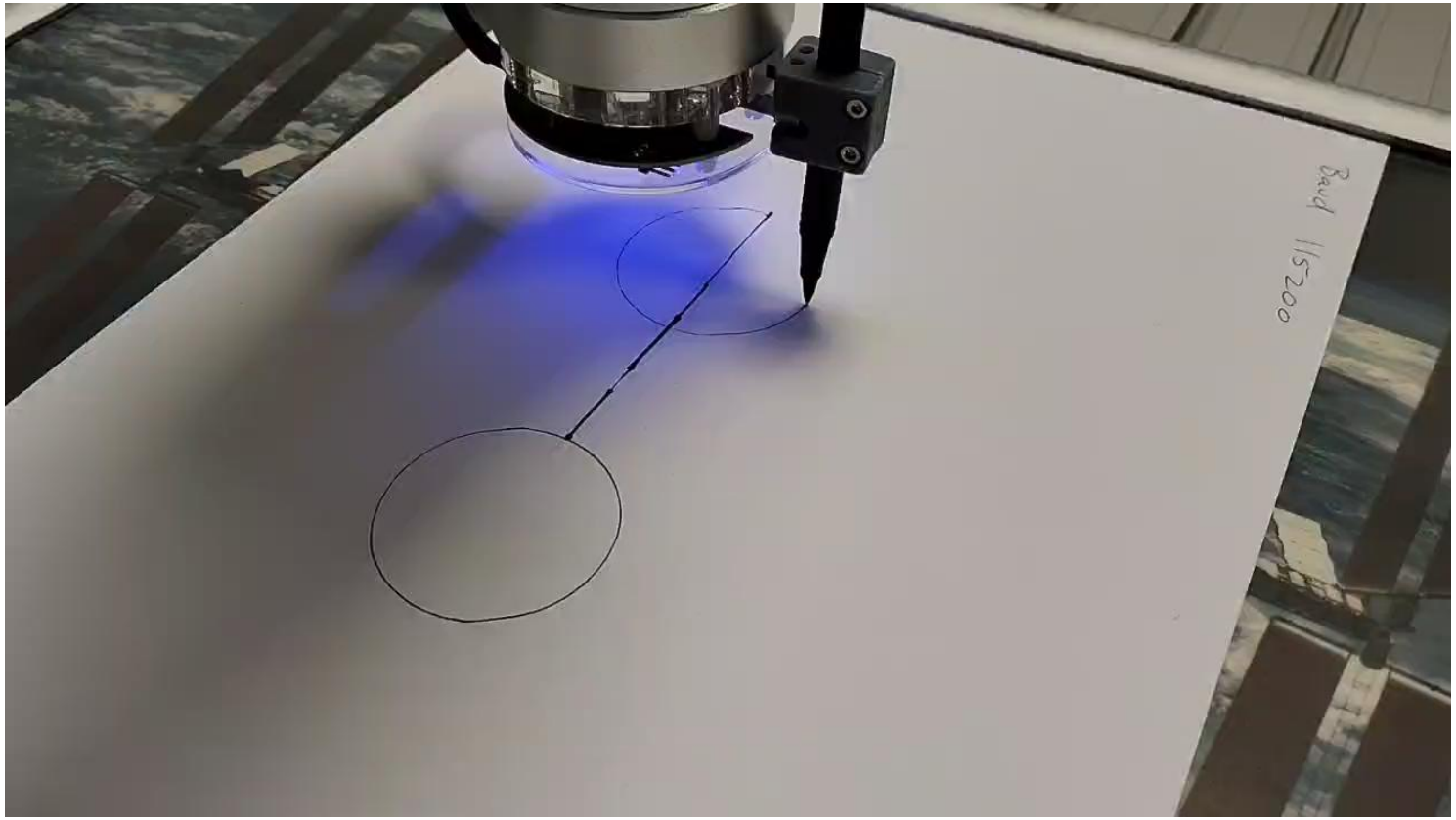
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Admittance control



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Your next assessment



This is one of the two tasks

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RICE: Reactive Interaction Controller for Cluttered Canopy Environment

Nidhi Homey Parayil[1], Thierry Peynot[2] and Chris Lehnert[3]

***Abstract*— Robotic navigation in dense, cluttered environments such as agricultural canopies presents significant challenges due to physical and visual occlusion caused by leaves and branches. Traditional vision-based or model-dependent approaches often fail in these settings, where physical interaction without damaging foliage and branches is necessary to reach a target. We present a novel reactive controller that enables safe navigation for a robotic arm in a contact-rich, cluttered, deformable environment using end-effector position and real-time tactile feedback. Our proposed framework's interaction strategy is based on a trade-off between minimizing disturbance by maneuvering around obstacles and pushing through them to move towards the target. We show that over 35 trials in 3 experimental plant setups with an occluded target, the proposed controller successfully reached the target in all trials without breaking any branch and outperformed the state-of-the-art model-free controller in robustness and adaptability. This work lays the foundation for safe, adaptive interaction in cluttered, contact-rich deformable environments, enabling future agricultural tasks such as pruning and harvesting in plant canopies.**

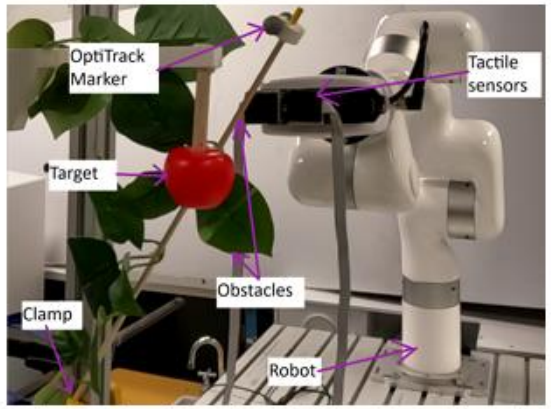


Fig. 1. Experimental setup: In a cluttered canopy, the robot navigates toward a given target, encountering branches and leaves as obstacles. OptiTrack markers measure branch disturbance. The robot adaptively pushes through or moves around obstacles to minimize disturbance using tactile feedback.

An Architecture for Reactive Mobile Manipulation *On-The-Move*

Ben Burgess-Limerick[1], Chris Lehnert[1], Jürgen Leitner[2], Peter Corke[1]

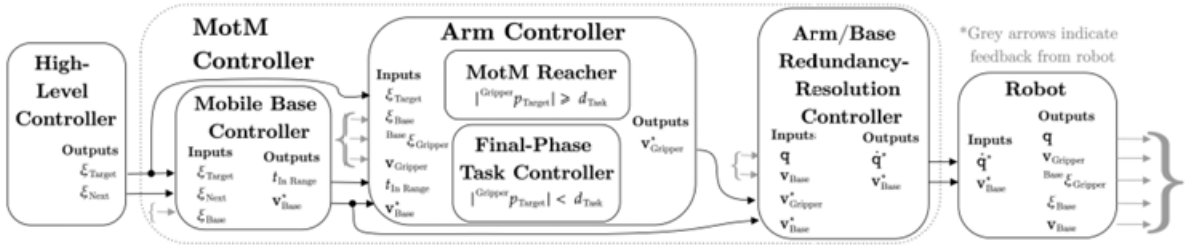
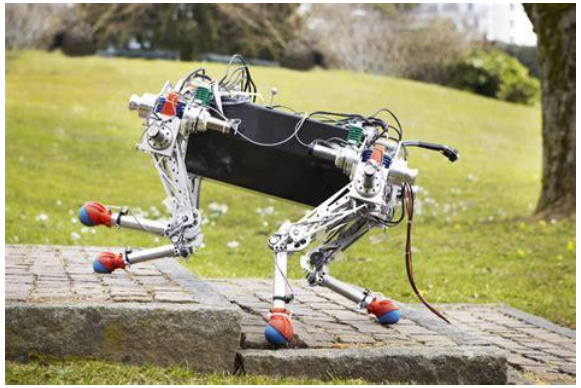


Fig. 2: Proposed architecture for reactive manipulation on-the-move. The arrows represent data flow between subsystems.

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Today's lecture

Kinematics of mobile robots



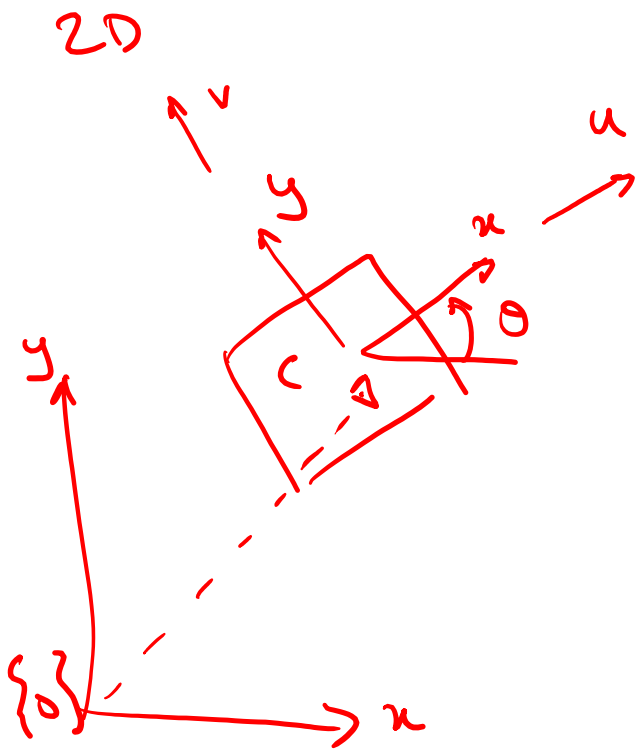
We will focus on wheeled robots, moving on flat ground

↳ Peter's book

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What is the kinematics problem in mobile robots?

In general, we are interested in the motion of the robot (i.e., most of it is velocity kinematics)



$$ {}^{0}V = \begin{Bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{Bmatrix}_{0} \qquad {}^{c}V = \begin{Bmatrix} u \\ v \\ \dot{\theta} \end{Bmatrix} $$

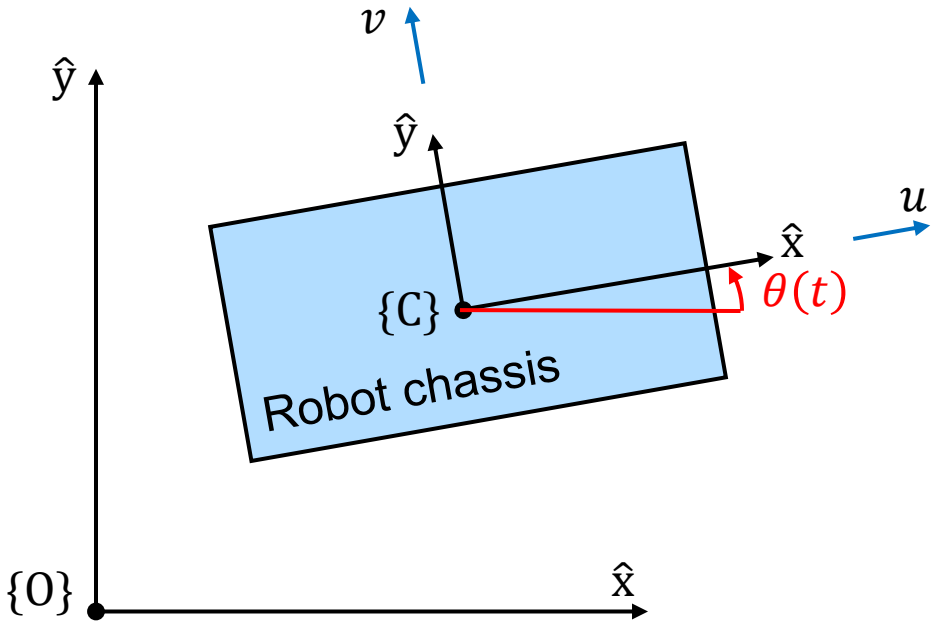
$$ \begin{Bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{Bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{Bmatrix} u \\ v \\ \dot{\theta} \end{Bmatrix} $$

J

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What is the kinematics problem in mobile robots?

In general, we are interested in the motion of the robot (i.e., most of it is velocity kinematics)



Let {O} be an inertial frame of reference and {C} be a frame attached to the robot.

The velocity of the robot represented in frame {O} is

$$^{O}v_{C} = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix}$$

The velocity of the robot represented in frame {C} is

$$^{C}v_{C} = \begin{bmatrix} u \\ v \\ \dot{\theta} \end{bmatrix}$$

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The two velocities are related as

chassis

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} u \\ v \\ \dot{\theta} \end{bmatrix}$$

zeta

$$\nu = \mathbf{J}(\theta)\zeta$$

Derivatives of generalised coordinates of the robot

Velocity in the robot frame

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Forward differential kinematics

$$\nu = \mathbf{J}(\theta)\zeta$$

Given an input velocity to the robot, how does it move in the inertial frame of reference?

Useful for simulation

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Inverse differential kinematics

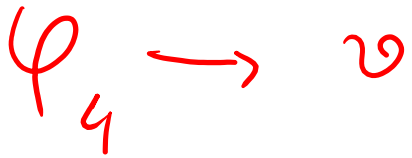
$$\zeta = \mathbf{J}^{-1}(\theta)\nu$$

What are the velocity commands to give to the robot for a desired motion in the inertial frame?

Useful for control

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How do we give input velocities to the robot?



- One cannot simply prescribe u, v and $\dot{\theta}$
- In a wheeled robot we control the velocities of the wheels.
- What is the relation between wheel velocities and chassis velocity?

$$\zeta = \mathbf{W}\dot{\varphi}$$

Velocity of the chassis

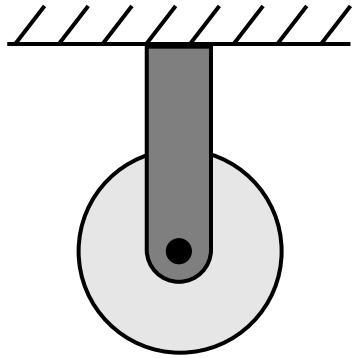
Velocity of the wheels

What does this term depend on?

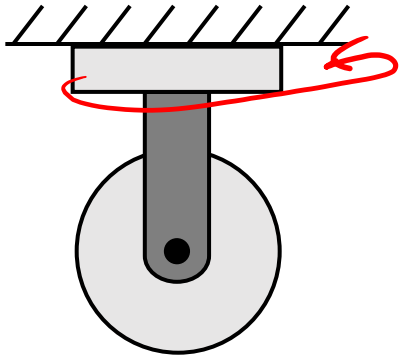
It depends on the configuration of the wheels

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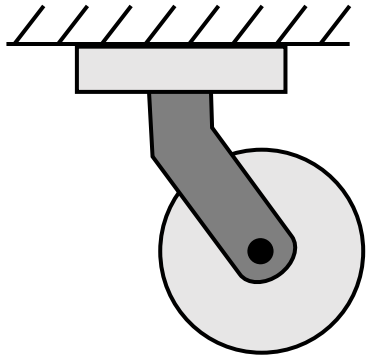
Types of wheels



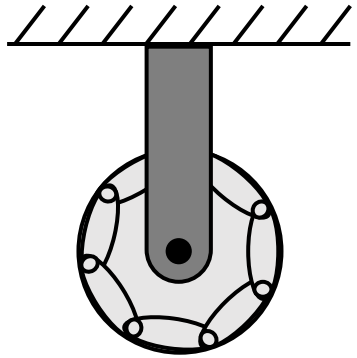
Fixed standard wheel



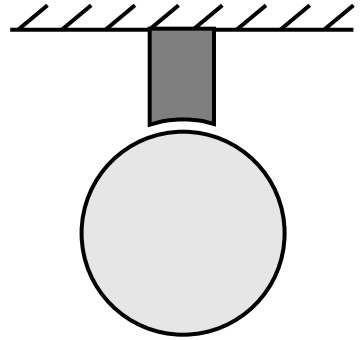
Steerable standard wheel



Castor wheel



Swedish wheel



Spherical wheel











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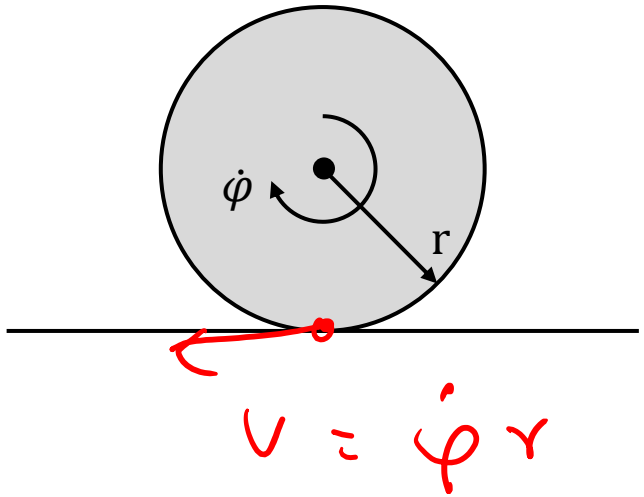
Wheels impose kinematic constraints

- Swedish wheels, spherical balls, and castors do not impose any constraint on the motion of the chassis.
- Only fixed standard wheels and steerable standard wheels impose constraints.
- In these two cases, lateral motion is not possible.

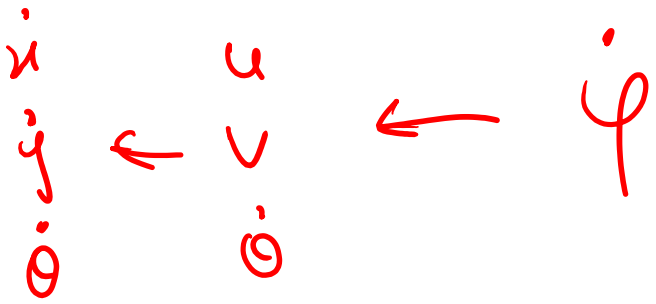
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We assume that all wheels exhibit pure-rolling

- In other words, there is no slipping at the point of contact of the wheel with the ground.



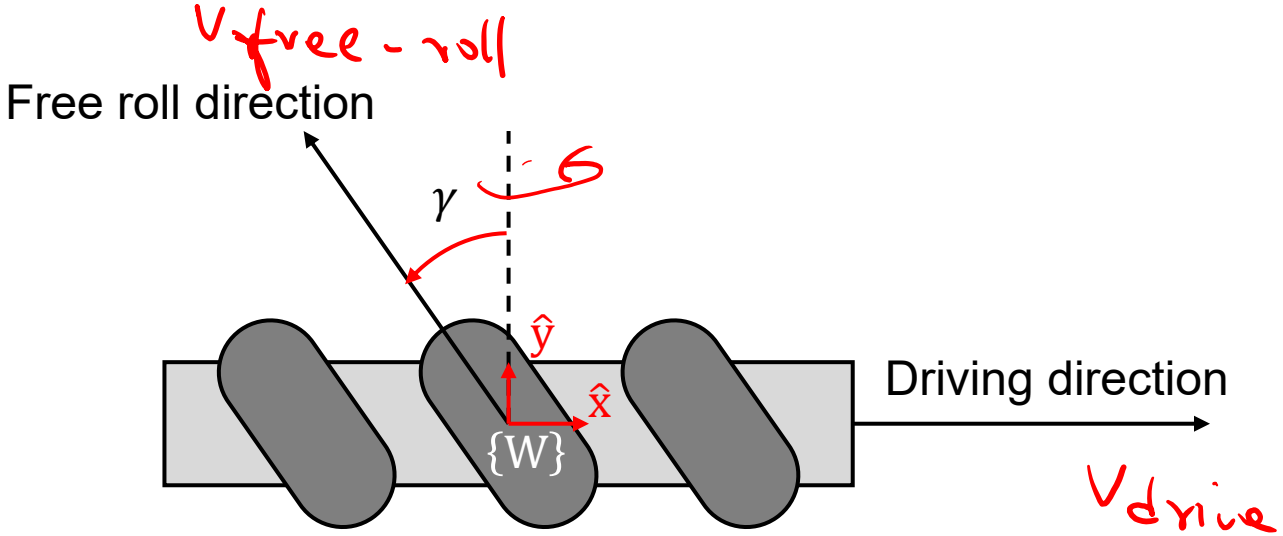
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Generalised wheel model

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Start with an omni-directional wheel



$v_{free-roll}$

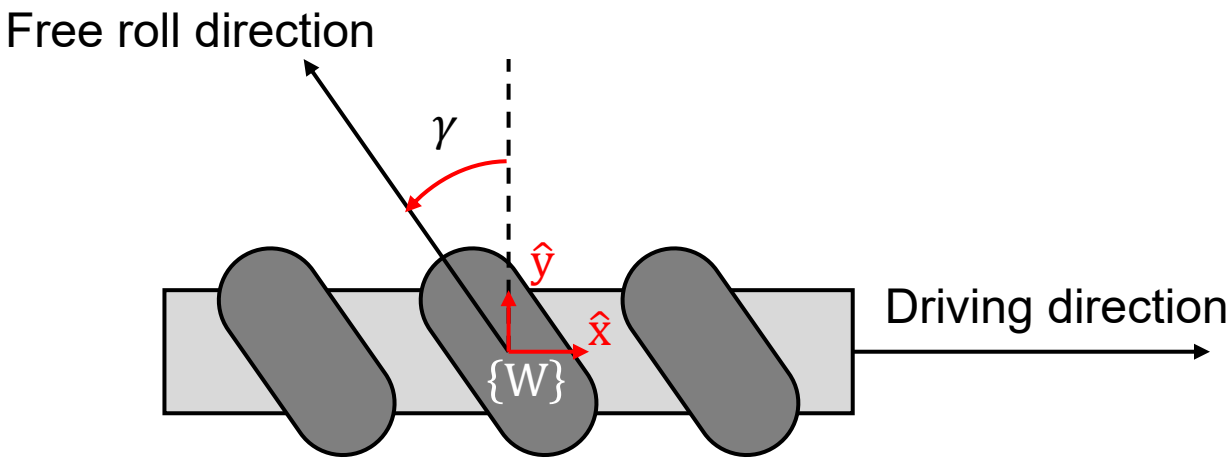
v_{drive}

$${}^{w}v_x = (1)v_{drive} + (-\sin\gamma)v_{free-roll}$$

$${}^{w}v_y = (0)v_{drive} + (\cos\gamma)v_{free-roll}$$

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Start with an omni-directional wheel



$$\begin{bmatrix} v_x \\ v_y \end{bmatrix} = v_{drive} \begin{bmatrix} 1 \\ 0 \end{bmatrix} + v_{free-roll} \begin{bmatrix} -\sin\gamma \\ \cos\gamma \end{bmatrix}$$

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Omni-directional wheel

$$v_x = 1\, v_{drive} - \sin\gamma\, v_{free}$$

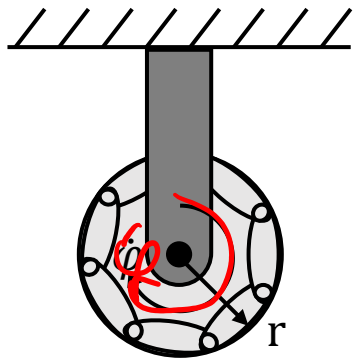
$$v_y = 0\, v_{drive} + \cos\gamma\, v_{free}$$

$$\dot{\varphi} = \frac{v_{drive}}{r} = \frac{v_x + \sin\gamma\, v_{free}}{r} = \frac{v_x + \frac{\sin\gamma}{\cos\gamma} v_y}{r}$$

$$\dot{\varphi} = \frac{v_x + \tan\gamma\, v_y}{r}$$

$$\dot{\varphi} = \begin{bmatrix} \frac{1}{r} & \frac{\tan\gamma}{r} \end{bmatrix} \begin{bmatrix} v_x \\ v_y \end{bmatrix}$$

$v_{drive} = r\dot{\varphi}$



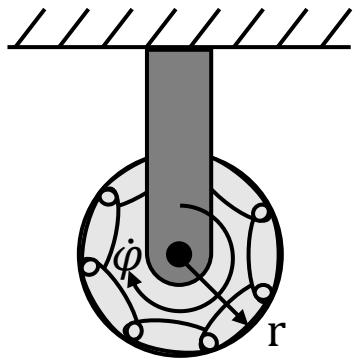
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Omni-directional wheel

$$\begin{bmatrix} v_x \\ v_y \end{bmatrix} = v_{drive} \begin{bmatrix} 1 \\ 0 \end{bmatrix} + v_{free-roll} \begin{bmatrix} -\sin\gamma \\ \cos\gamma \end{bmatrix}$$

$$v_{drive} = v_x + v_y \tan\gamma$$

$$v_{free-roll} = \frac{v_y}{\cos\gamma}$$

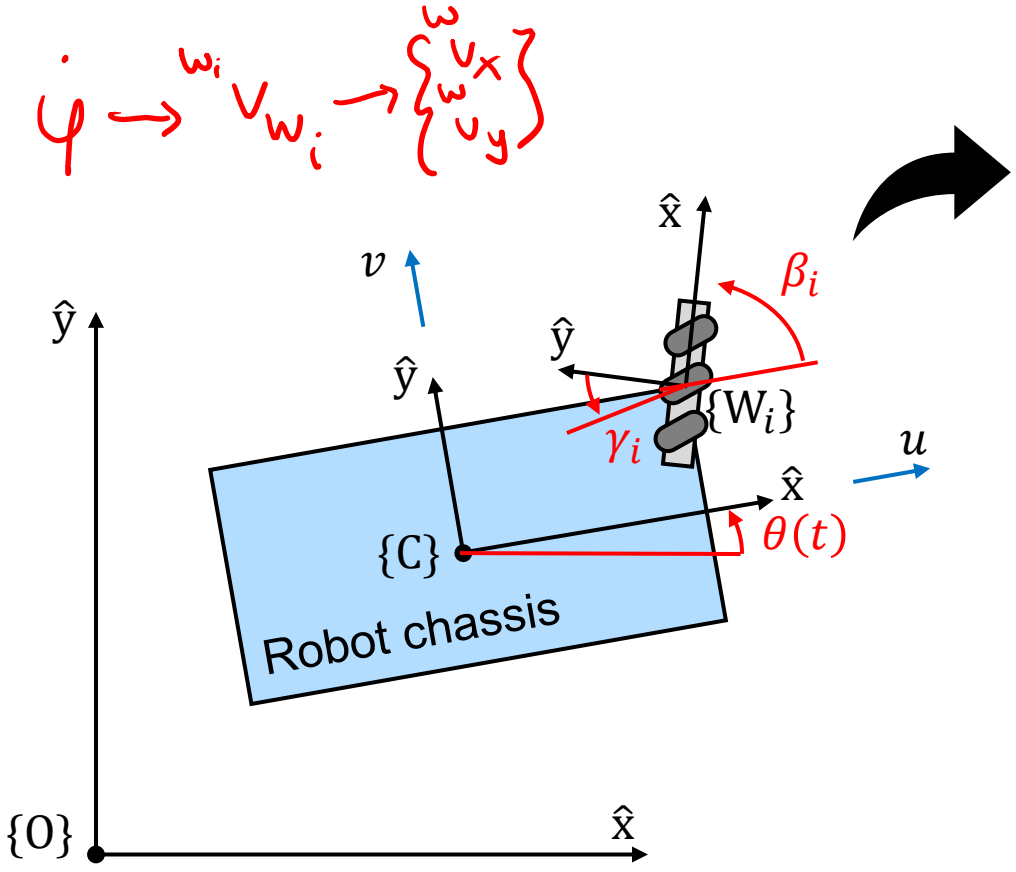


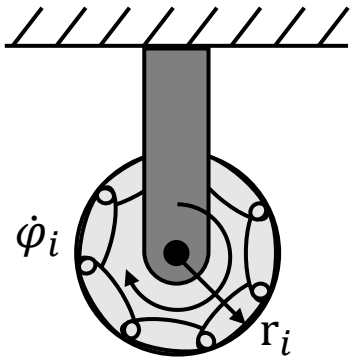
How are these terms related to the angular velocity of the wheel?

$$\dot{\varphi} = \frac{v_{drive}}{r} = \frac{v_x + v_y \tan\gamma}{r} = \begin{bmatrix} \frac{1}{r_i} & \frac{\tan\gamma_i}{r_i} \end{bmatrix} \begin{bmatrix} v_x \\ v_y \end{bmatrix}$$

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This wheel is fixed to the chassis of the robot



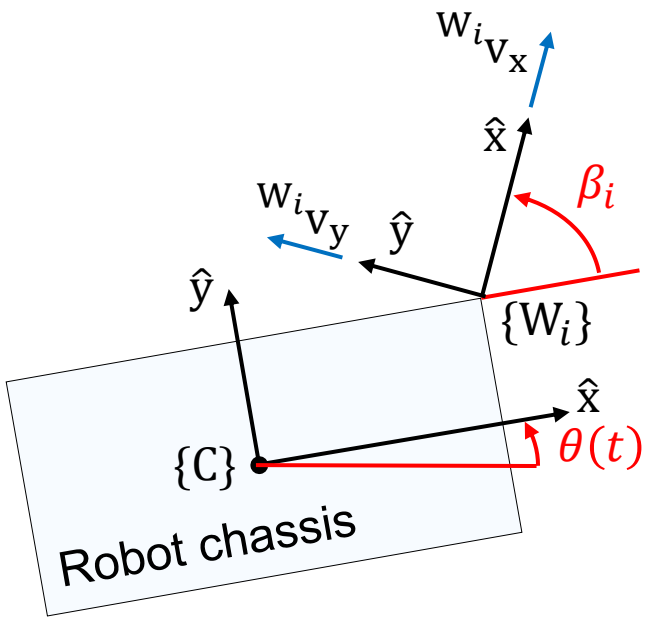


We want to determine the relation between angular velocity of the wheel, $\dot{\varphi}_i$

and the robot's spatial velocity, ${}^{O}v_C = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix}$

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Step 1

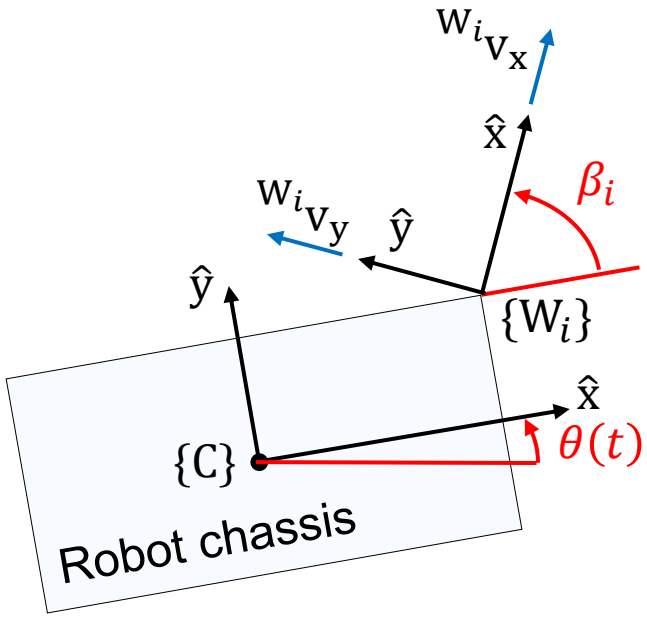


${}^{C}v_{w} \leftarrow {}^{w}v_{w}$?

Step 1: Express the velocity of the wheel $\begin{bmatrix} {}^{w_i}v_x \\ {}^{w_i}v_y \end{bmatrix}$

in the frame of reference of the chassis {C}

Step 1



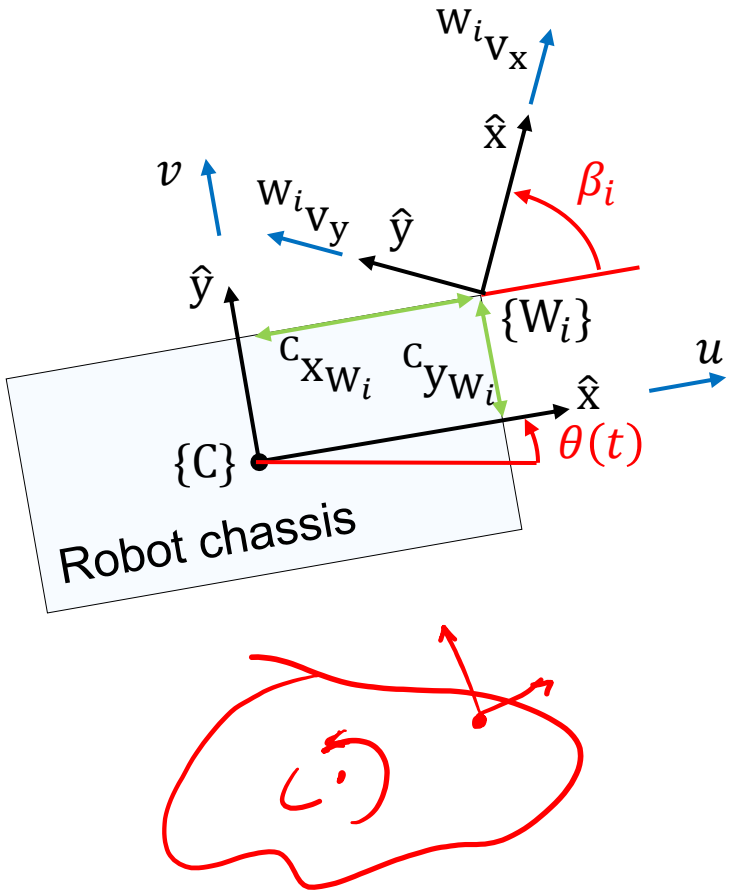
Step 1: Express the velocity of the wheel ${}^{W_i}v_{W_i} = \begin{bmatrix} {}^{W_i}v_x \\ {}^{W_i}v_y \end{bmatrix}$

in the frame of reference of the chassis {C}

$${}^{C}v_{W_i} = \begin{bmatrix} \cos\beta_i & -\sin\beta_i \\ \sin\beta_i & \cos\beta_i \end{bmatrix} \begin{bmatrix} {}^{W_i}v_x \\ {}^{W_i}v_y \end{bmatrix}$$

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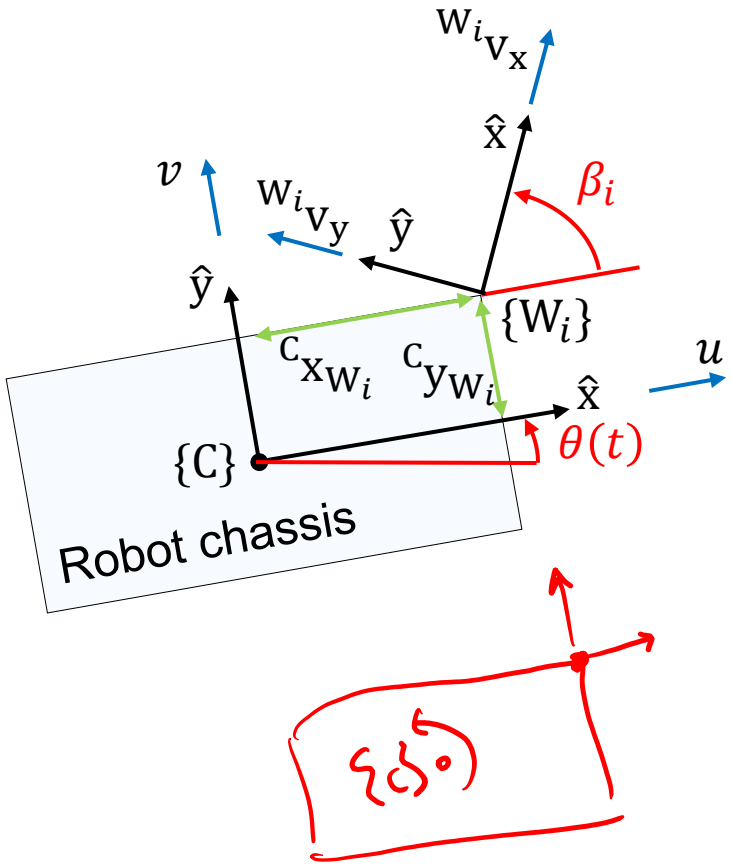
Step 2



Step 2: Express the velocity of the wheel in terms of the velocity of the robot chassis ${}^{C}v_C = \begin{bmatrix} u \\ v \\ \dot{\theta} \end{bmatrix}$

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Step 2



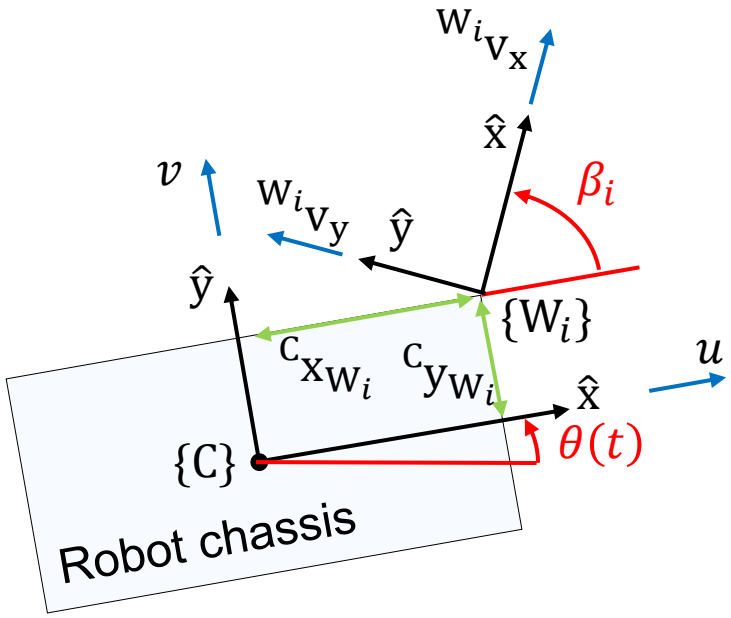
Step 2: Express the velocity of the wheel in terms of the velocity of the robot chassis ${}^{C}v_C = \begin{bmatrix} u \\ v \\ \dot{\theta} \end{bmatrix}$

Since {C} is rotating with angular velocity $\dot{\theta}$, we have the relation

$${}^{C}v_{W_i} = \begin{bmatrix} u - \dot{\theta}\,{}^{C}y_{W_i} \\ v + \dot{\theta}\,{}^{C}x_{W_i} \end{bmatrix} = \begin{bmatrix} 1 & 0 & -{}^{C}y_{W_i} \\ 0 & 1 & {}^{C}x_{W_i} \end{bmatrix} \begin{bmatrix} u \\ v \\ \dot{\theta} \end{bmatrix}$$

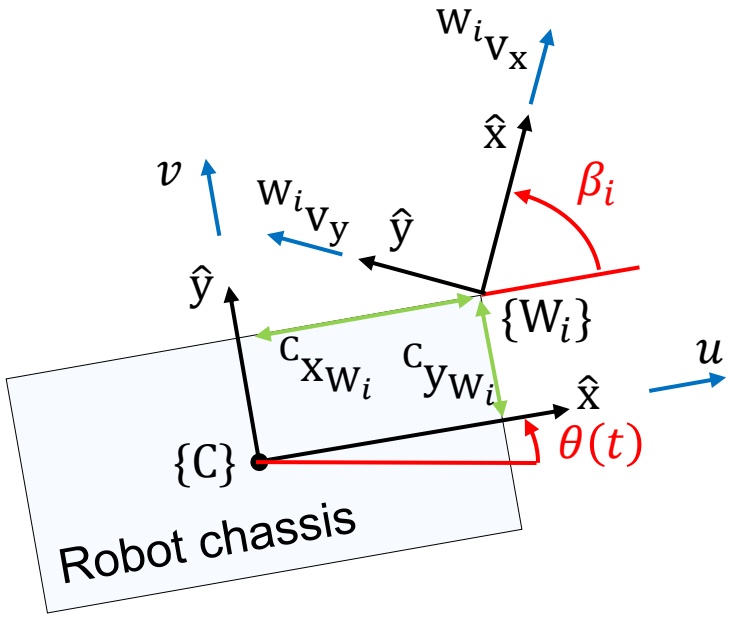
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From step 1 and step 2



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From step 1 and step 2



$$ {}^{C}v_{W_i} = \begin{bmatrix} \cos\beta_i & -\sin\beta_i \\ \sin\beta_i & \cos\beta_i \end{bmatrix} \begin{bmatrix} {}^{W_i}v_x \\ {}^{W_i}v_y \end{bmatrix} $$ ← From step 1

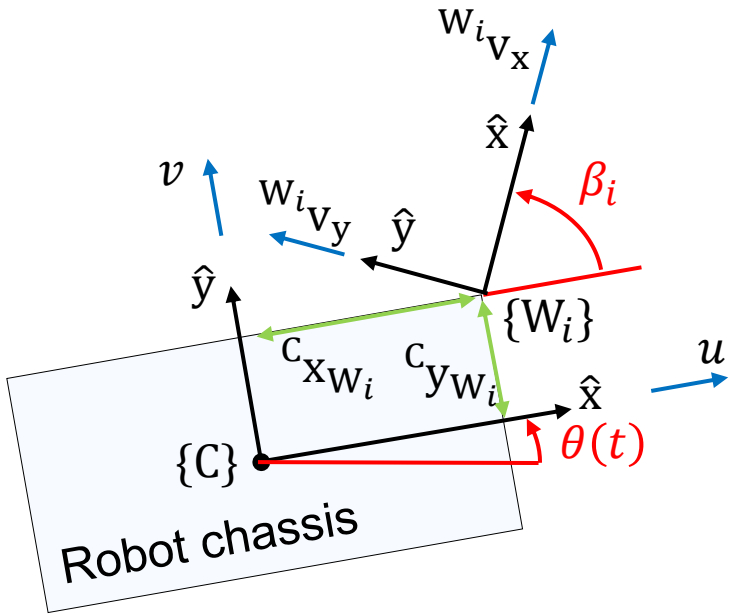
$$ {}^{C}v_{W_i} = \begin{bmatrix} u - \dot{\theta}\ {}^{C}y_{W_i} \\ v + \dot{\theta}\ {}^{C}x_{W_i} \end{bmatrix} = \begin{bmatrix} 1 & 0 & -{}^{C}y_{W_i} \\ 0 & 1 & {}^{C}x_{W_i} \end{bmatrix} \begin{bmatrix} u \\ v \\ \dot{\theta} \end{bmatrix} $$ ← From step 2

Therefore,

$$ \begin{bmatrix} \cos\beta_i & -\sin\beta_i \\ \sin\beta_i & \cos\beta_i \end{bmatrix} \begin{bmatrix} {}^{W_i}v_x \\ {}^{W_i}v_y \end{bmatrix} = \begin{bmatrix} 1 & 0 & -{}^{C}y_{W_i} \\ 0 & 1 & {}^{C}x_{W_i} \end{bmatrix} \begin{bmatrix} u \\ v \\ \dot{\theta} \end{bmatrix} $$

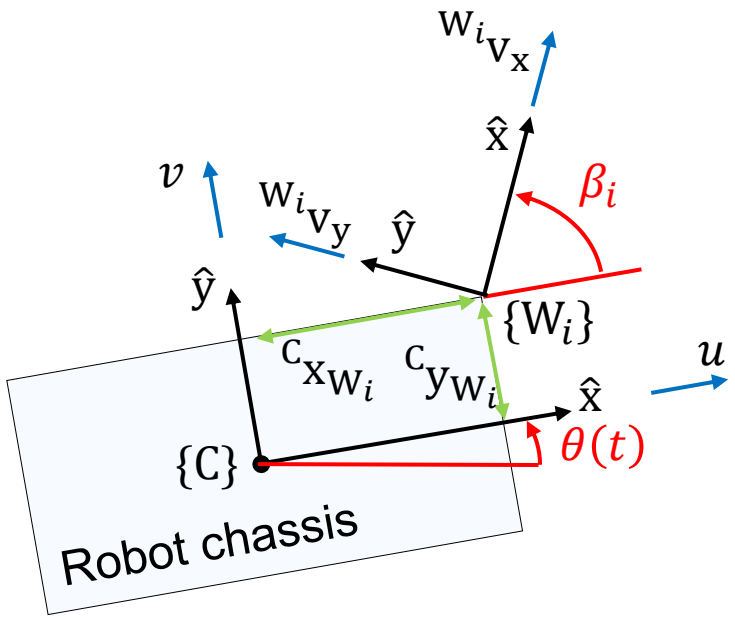
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Obtain expression for $\dot{\varphi}_i$



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Obtain expression for $\dot{\varphi}_i$



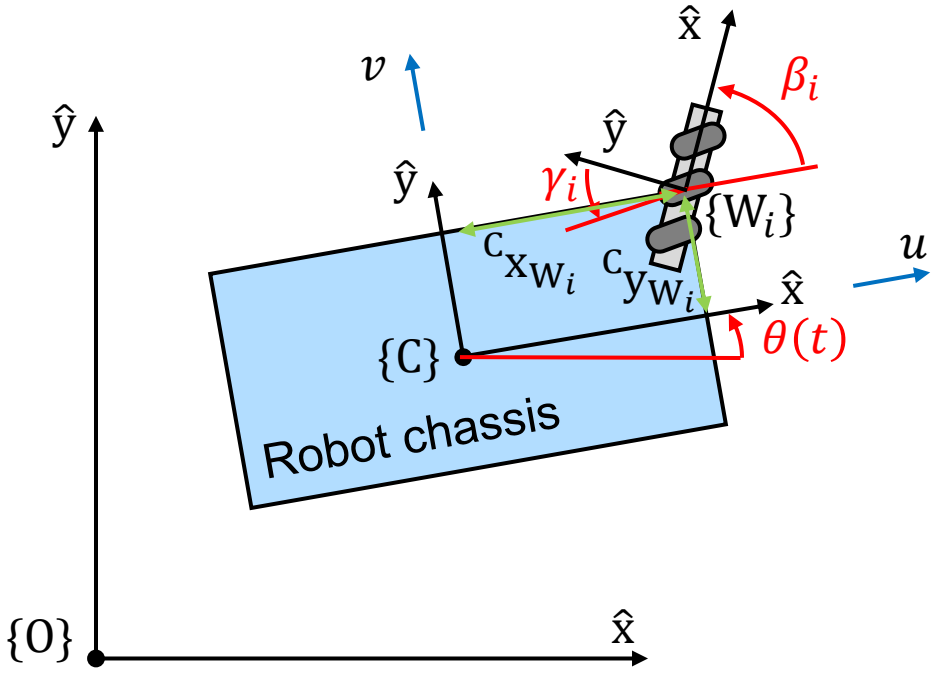
$$\begin{bmatrix} \cos\beta_i & -\sin\beta_i \\ \sin\beta_i & \cos\beta_i \end{bmatrix} \begin{bmatrix} {}^{w_i}v_x \\ {}^{w_i}v_y \end{bmatrix} = \begin{bmatrix} 1 & 0 & -{}^{c}y_{w_i} \\ 0 & 1 & {}^{c}x_{w_i} \end{bmatrix} \begin{bmatrix} u \\ v \\ \dot{\theta} \end{bmatrix}$$

$$\begin{bmatrix} {}^{w_i}v_x \\ {}^{w_i}v_y \end{bmatrix} = \begin{bmatrix} \cos\beta_i & \sin\beta_i \\ -\sin\beta_i & \cos\beta_i \end{bmatrix} \begin{bmatrix} 1 & 0 & -{}^{c}y_{w_i} \\ 0 & 1 & {}^{c}x_{w_i} \end{bmatrix} \begin{bmatrix} u \\ v \\ \dot{\theta} \end{bmatrix}$$

$$\dot{\varphi}_i = \begin{bmatrix} \frac{1}{r_i} & \frac{\tan\gamma_i}{r_i} \end{bmatrix} \begin{bmatrix} {}^{w_i}v_x \\ {}^{w_i}v_y \end{bmatrix} = \begin{bmatrix} \frac{1}{r_i} & \frac{\tan\gamma_i}{r_i} \end{bmatrix} \begin{bmatrix} \cos\beta_i & \sin\beta_i \\ -\sin\beta_i & \cos\beta_i \end{bmatrix} \begin{bmatrix} 1 & 0 & -{}^{c}y_{w_i} \\ 0 & 1 & {}^{c}x_{w_i} \end{bmatrix} \begin{bmatrix} u \\ v \\ \dot{\theta} \end{bmatrix}$$

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If one is observing from a fixed inertial frame {O}



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The generalised model of a wheeled robot

$$\dot{\varphi}_i = \begin{bmatrix} \frac{1}{r_i} & \frac{\tan\gamma_i}{r_i} \end{bmatrix} \begin{bmatrix} \cos\beta_i & \sin\beta_i \\ -\sin\beta_i & \cos\beta_i \end{bmatrix} \begin{bmatrix} 1 & 0 & -{}^{C}y_{W_i} \\ 0 & 1 & {}^{C}x_{W_i} \end{bmatrix} \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix}$$

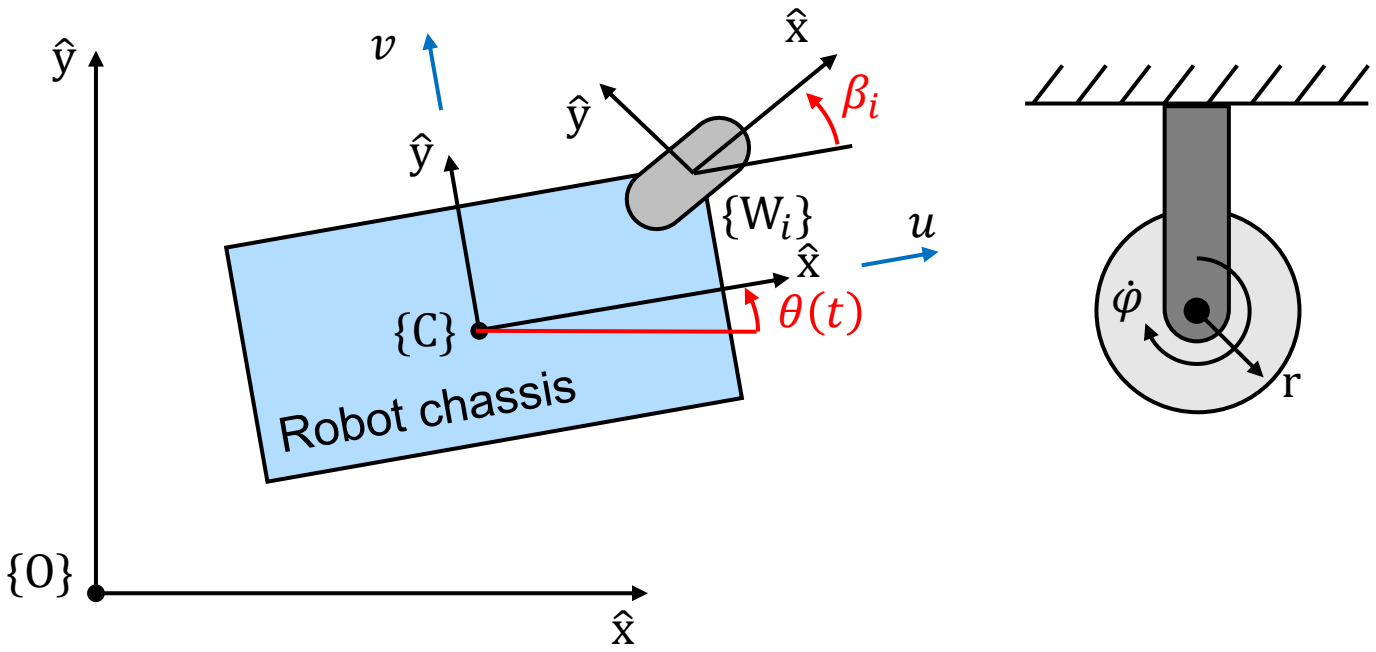
$\dot{\varphi}_i$: the angular velocity of the wheel
r_i : the radius of the wheel
γ_i : the angle between $v_{free-roll}$ and the normal to the plane of the wheel
β_i : the angle between the v_{drive} and the robot chassis
${}^{C}x_{W_i}$ and ${}^{C}y_{W_i}$: the position of the wheel with respect to the chassis
θ : the orientation of the robot chassis with respect to a global inertial frame of reference
$\dot{x}$, $\dot{y}$, $\dot{\theta}$: the velocity of the robot chassis in the global inertial frame of reference

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Robot with a fixed standard wheel

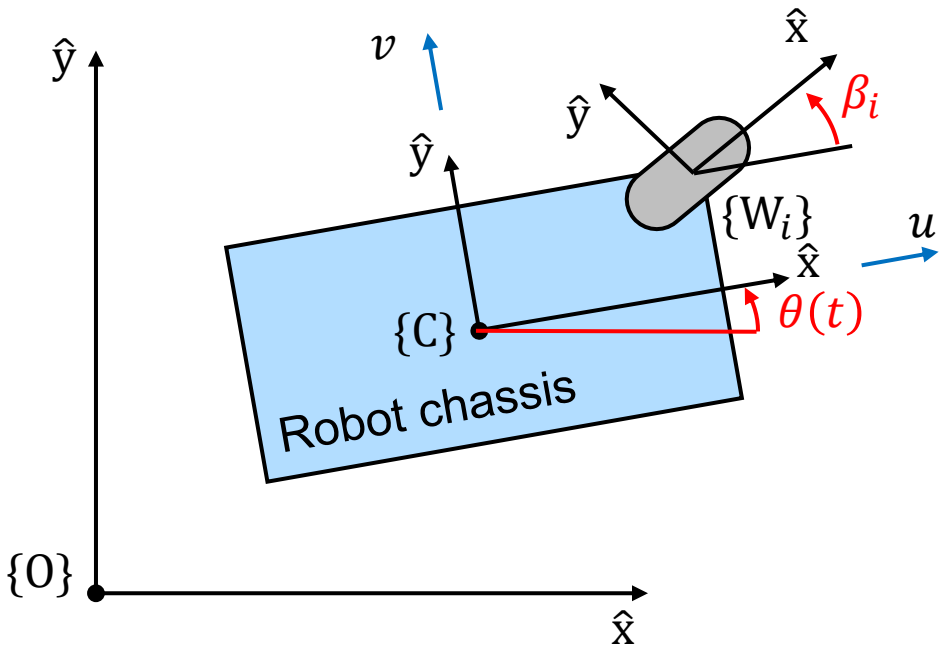
TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

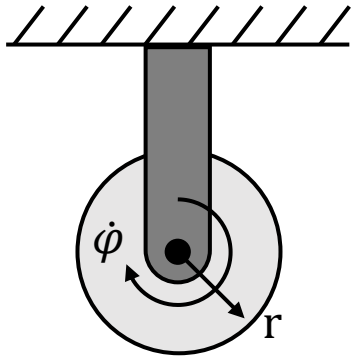
Robot with fixed wheel



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Robot with fixed wheel





$\gamma_i = 0$ → Technically, gamma is not defined. This wheel cannot slide sideways.

β_i = constant

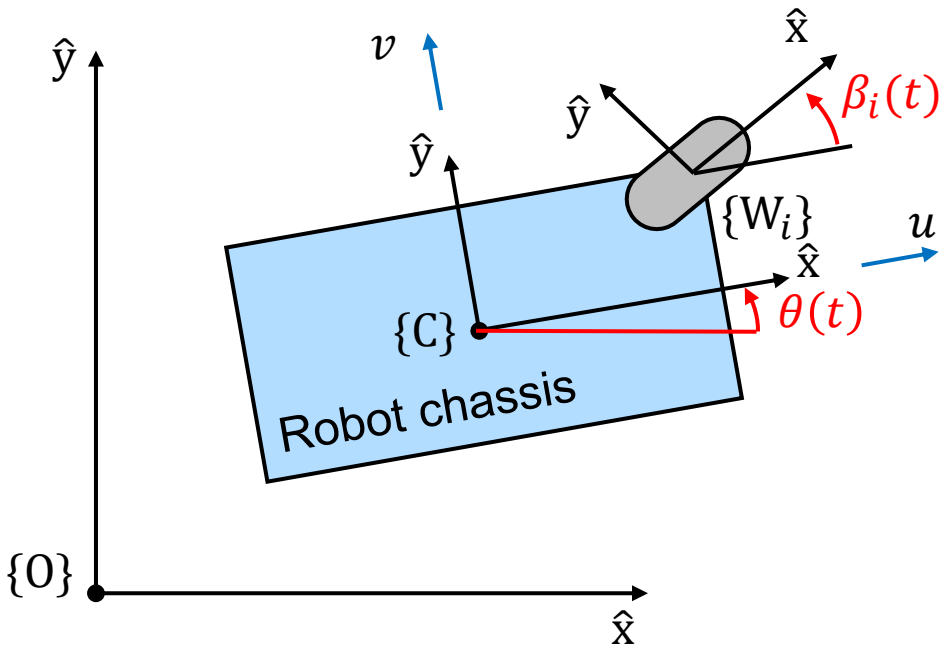
$$\dot{\varphi}_i = \begin{bmatrix} \frac{1}{r_i} & 0 \end{bmatrix} \begin{bmatrix} \cos\beta_i & \sin\beta_i \\ -\sin\beta_i & \cos\beta_i \end{bmatrix} \begin{bmatrix} 1 & 0 & -{}^{c}y_{W_i} \\ 0 & 1 & {}^{c}x_{W_i} \end{bmatrix} \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix}$$

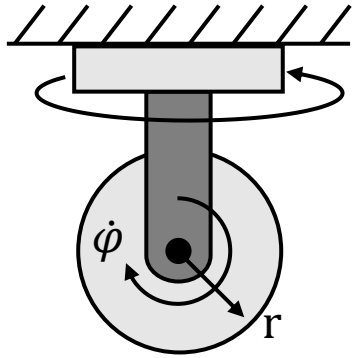
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Robot with a steerable standard wheel

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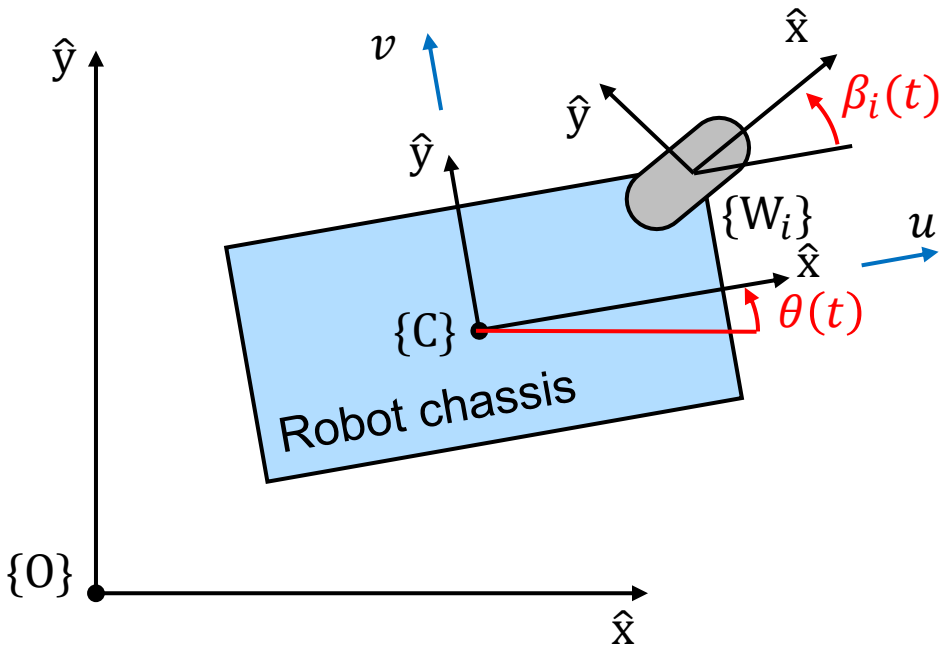
Robot with a steerable wheel

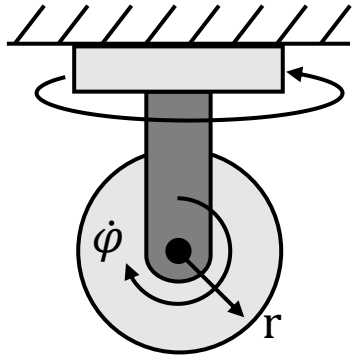




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Robot with a steerable wheel





$\gamma_i = 0$ → Technically, gamma is not defined. This wheel cannot slide sideways.

β_i is a function of time

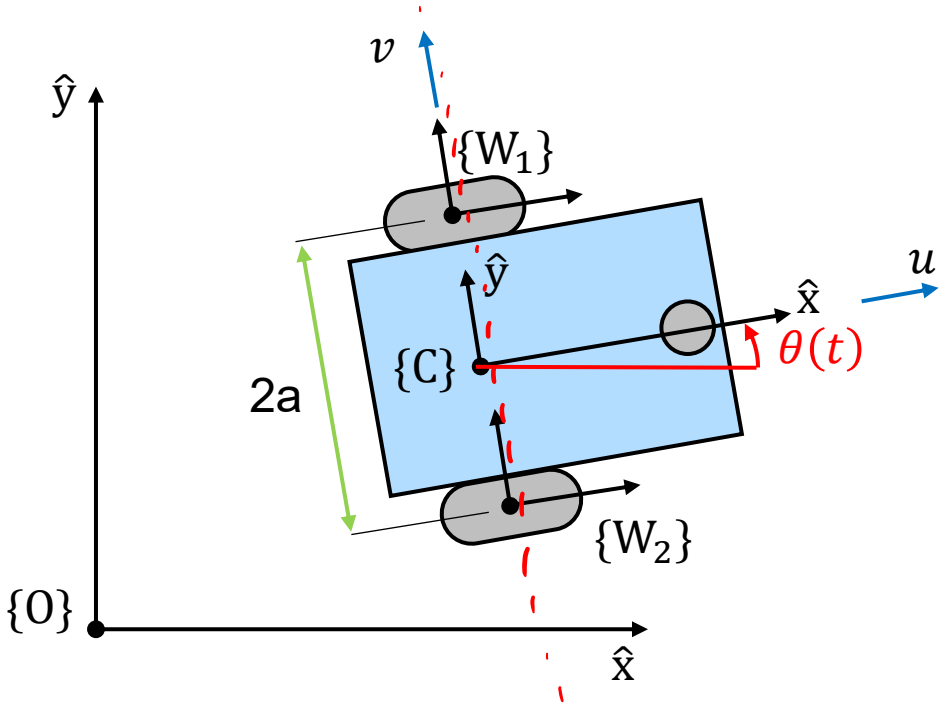
$$\dot{\varphi}_i = \begin{bmatrix} \frac{1}{r_i} & 0 \end{bmatrix} \begin{bmatrix} \cos\beta_i & \sin\beta_i \\ -\sin\beta_i & \cos\beta_i \end{bmatrix} \begin{bmatrix} 1 & 0 & -{}^{c}y_{W_i} \\ 0 & 1 & {}^{c}x_{W_i} \end{bmatrix} \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix}$$

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Differential-drive robot

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Differential drive robot



r_1 : r r

γ : 0 0

β : 0 0

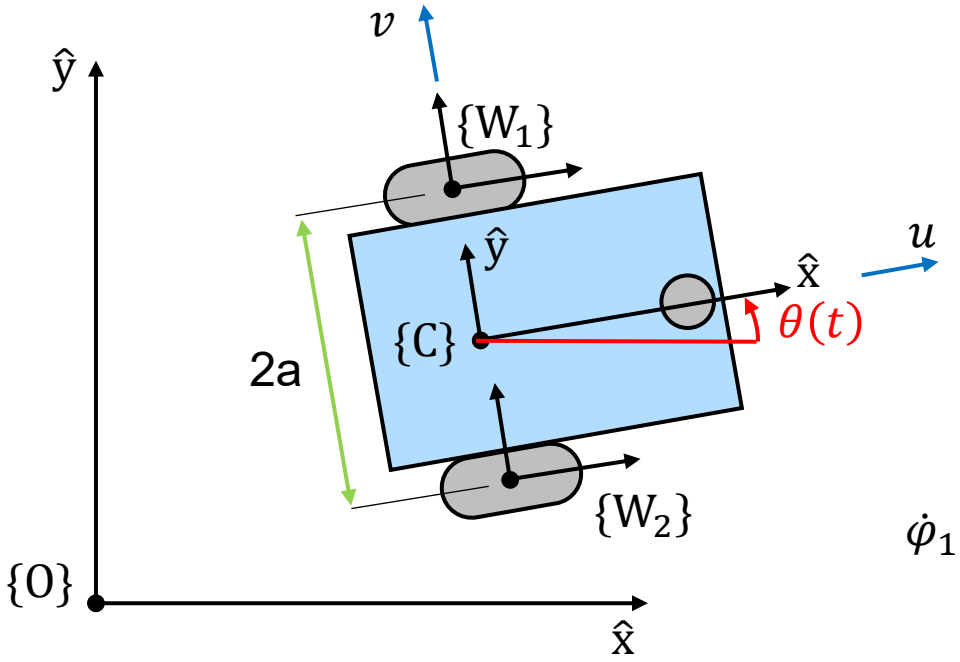
c_{x_w} : 0 0

c_{y_w} : a $-a$

Standard wheels with no rollers.
Wheel radius is r.
Each wheel is independently actuated.

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Differential drive robot



$$r_1 = r$$
$$\gamma_1 = 0$$
$$\beta_1 = 0$$
$$c_{x_{W_1}} = 0$$
$$c_{y_{W_1}} = a$$

$$\dot{\varphi}_1 = \begin{bmatrix} \frac{1}{r} & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -a \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} u \\ v \\ \dot{\theta} \end{bmatrix}$$

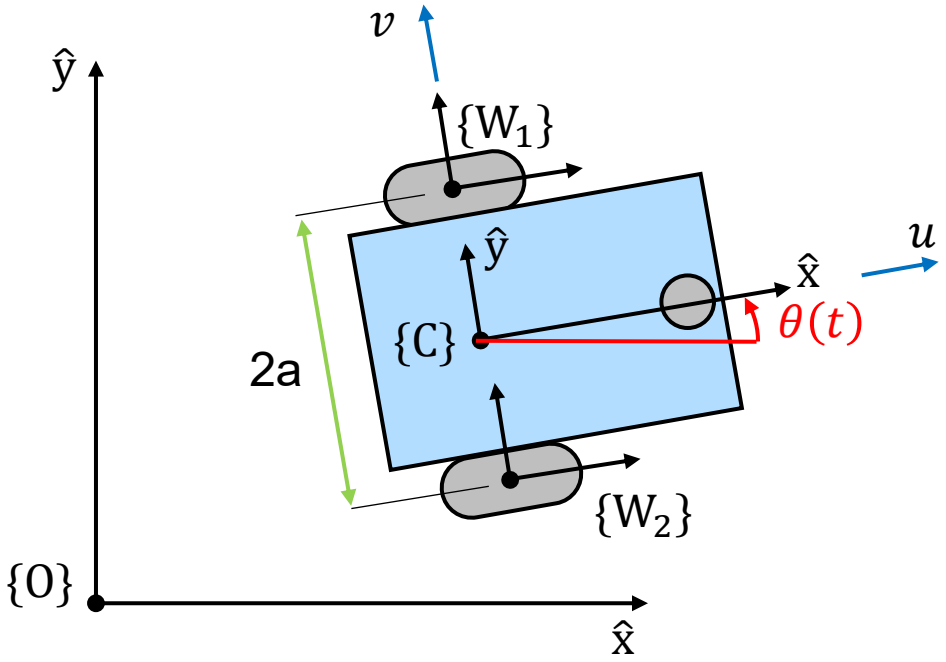
$$r_2 = r$$
$$\gamma_2 = 0$$
$$\beta_2 = 0$$
$$c_{x_{W_2}} = 0$$
$$c_{y_{W_2}} = -a$$

$$\dot{\varphi}_2 = \begin{bmatrix} \frac{1}{r} & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & a \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} u \\ v \\ \dot{\theta} \end{bmatrix}$$

Standard wheels with no rollers.
Wheel radius is r.
Each wheel is independently actuated.

TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

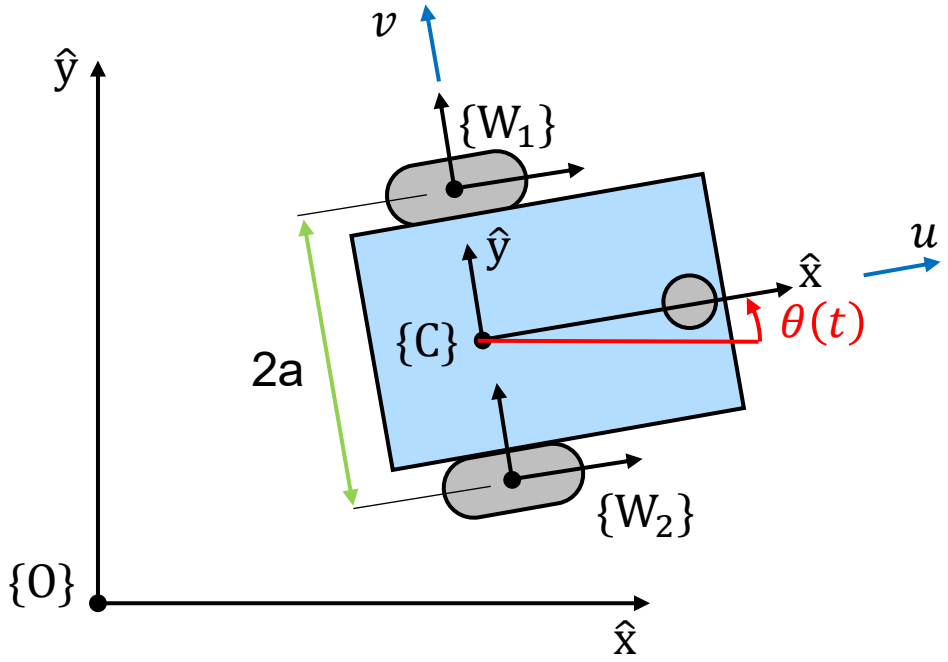
Differential drive robot



Standard wheels with no rollers.
Wheel radius is r.
Each wheel is independently actuated.

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Differential drive robot



$$\begin{bmatrix}\dot{\varphi}_1\\ \dot{\varphi}_2\end{bmatrix} = \begin{bmatrix}\frac{1}{r} & 0 & -\frac{a}{r}\\ \frac{1}{r} & 0 & \frac{a}{r}\end{bmatrix}\begin{bmatrix}u\\ v\\ \dot{\theta}\end{bmatrix}$$

2 × 3

$$\begin{bmatrix}u\\ v\\ \dot{\theta}\end{bmatrix} = \begin{bmatrix}\frac{r}{2} & \frac{r}{2}\\ 0 & 0\\ \frac{-r}{2a} & \frac{r}{2a}\end{bmatrix}\begin{bmatrix}\dot{\varphi}_1\\ \dot{\varphi}_2\end{bmatrix}$$

3 × 2

$$\zeta = \mathbf{W}\dot{\varphi}$$

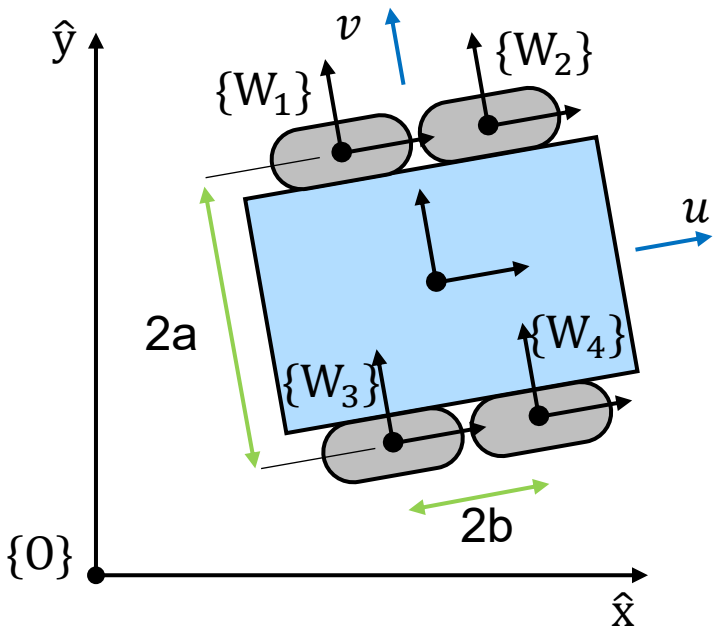
Standard wheels with no rollers.
Wheel radius is r.
Each wheel is independently actuated.

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Skid-steer robot

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Skid-steer robot

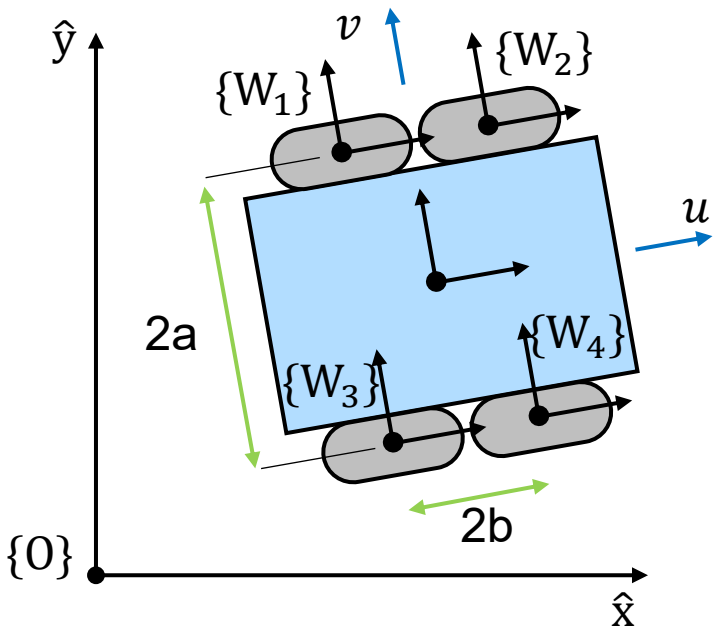


	w_1	w_2	w_3	w_4
r	r	r	r	r
γ	0	0	0	0
β	0	0	0	0
c_{x_w}	$-b$	b	$-b$	b
c_{y_w}	a	a	$-a$	$-a$

Standard wheels with no rollers.
Wheel radius is r.
Each wheel is independently actuated.

TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

Skid-steer robot



$$r_1 = r \quad \gamma_1 = 0 \quad \beta_1 = 0 \quad {}^{c}x_{W_1} = -b \quad {}^{c}y_{W_1} = a$$

$$r_2 = r \quad \gamma_2 = 0 \quad \beta_2 = 0 \quad {}^{c}x_{W_2} = b \quad {}^{c}y_{W_2} = a$$

$$r_3 = r \quad \gamma_3 = 0 \quad \beta_3 = 0 \quad {}^{c}x_{W_3} = -b \quad {}^{c}y_{W_3} = -a$$

$$r_4 = r \quad \gamma_4 = 0 \quad \beta_4 = 0 \quad {}^{c}x_{W_4} = b \quad {}^{c}y_{W_4} = -a$$

$$\dot{\varphi}_1 = \begin{bmatrix} \frac{1}{r} & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -a \\ 0 & 1 & -b \end{bmatrix} \begin{bmatrix} u \\ v \\ \dot{\theta} \end{bmatrix}$$

$$\dot{\varphi}_3 = \begin{bmatrix} \frac{1}{r} & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & a \\ 0 & 1 & -b \end{bmatrix} \begin{bmatrix} u \\ v \\ \dot{\theta} \end{bmatrix}$$

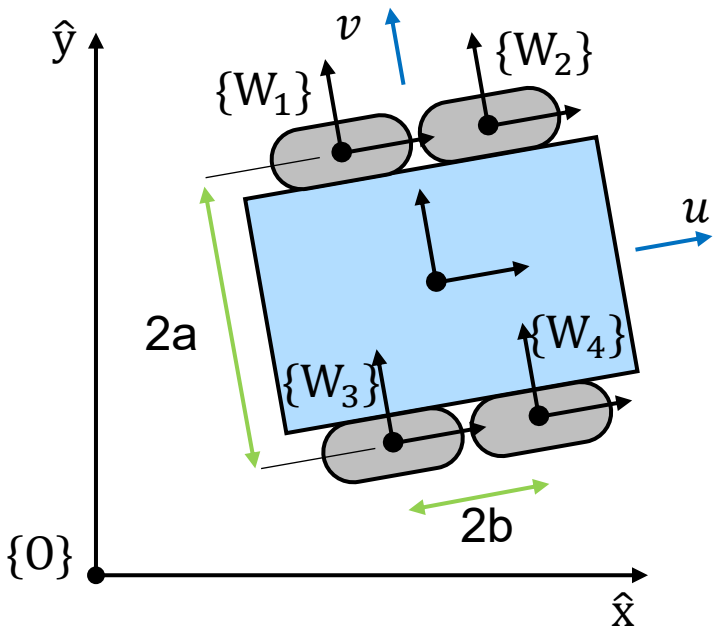
$$\dot{\varphi}_2 = \begin{bmatrix} \frac{1}{r} & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -a \\ 0 & 1 & b \end{bmatrix} \begin{bmatrix} u \\ v \\ \dot{\theta} \end{bmatrix}$$

$$\dot{\varphi}_4 = \begin{bmatrix} \frac{1}{r} & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & a \\ 0 & 1 & b \end{bmatrix} \begin{bmatrix} u \\ v \\ \dot{\theta} \end{bmatrix}$$

Standard wheels with no rollers.
Wheel radius is r.
Each wheel is independently actuated.

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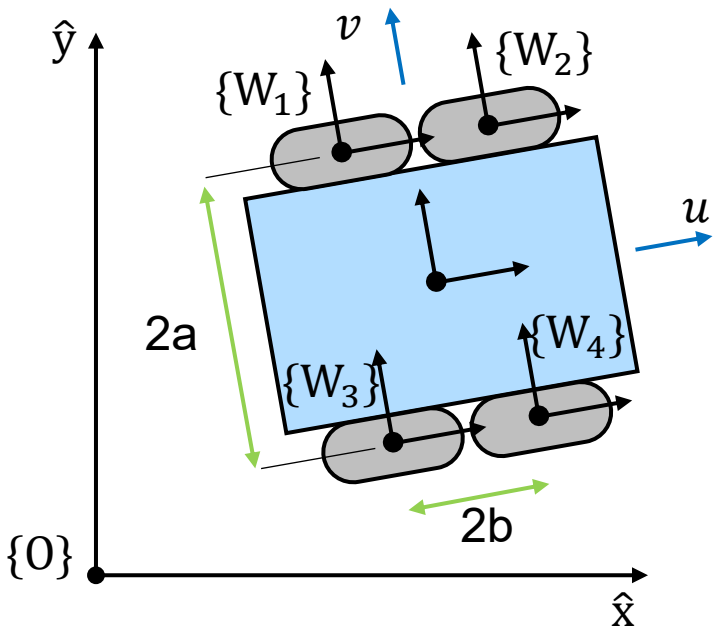
Skid-steer robot



Standard wheels with no rollers.
Wheel radius is r.
Each wheel is independently actuated.

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Skid-steer robot



$$\begin{bmatrix}\dot{\varphi}_1\\ \dot{\varphi}_2\\ \dot{\varphi}_3\\ \dot{\varphi}_4\end{bmatrix} = \begin{bmatrix}1/r & 0 & -a/r\\ 1/r & 0 & -a/r\\ 1/r & 0 & a/r\\ 1/r & 0 & a/r\end{bmatrix}\begin{bmatrix}u\\ v\\ \dot{\theta}\end{bmatrix}$$

$$\begin{bmatrix}u\\ v\\ \dot{\theta}\end{bmatrix} = \begin{bmatrix}r/4 & r/4 & r/4 & r/4\\ 0 & 0 & 0 & 0\\ -r/4a & -r/4a & r/4a & r/4a\end{bmatrix}\begin{bmatrix}\dot{\varphi}_1\\ \dot{\varphi}_2\\ \dot{\varphi}_3\\ \dot{\varphi}_4\end{bmatrix}$$

$$\zeta = \mathbf{W}\dot{\varphi}$$

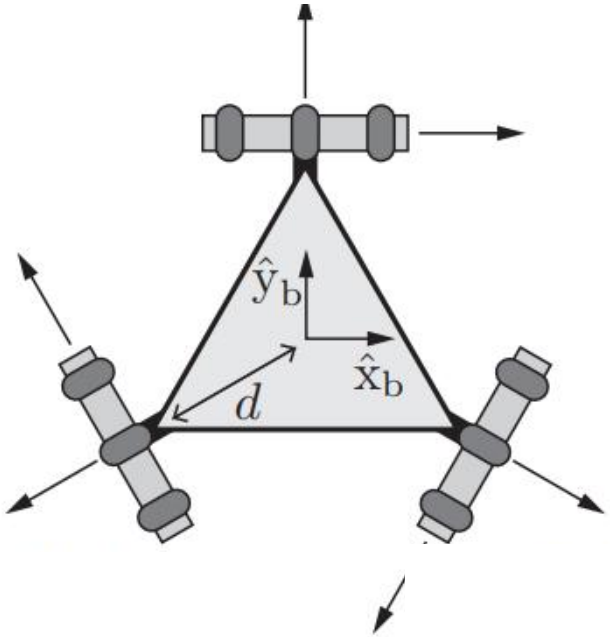
Standard wheels with no rollers.
Wheel radius is r.
Each wheel is independently actuated.

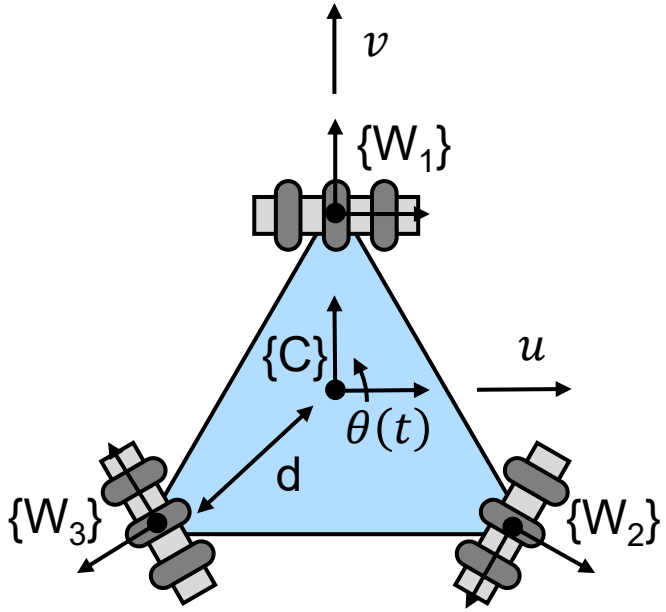
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Robots with Swedish wheels

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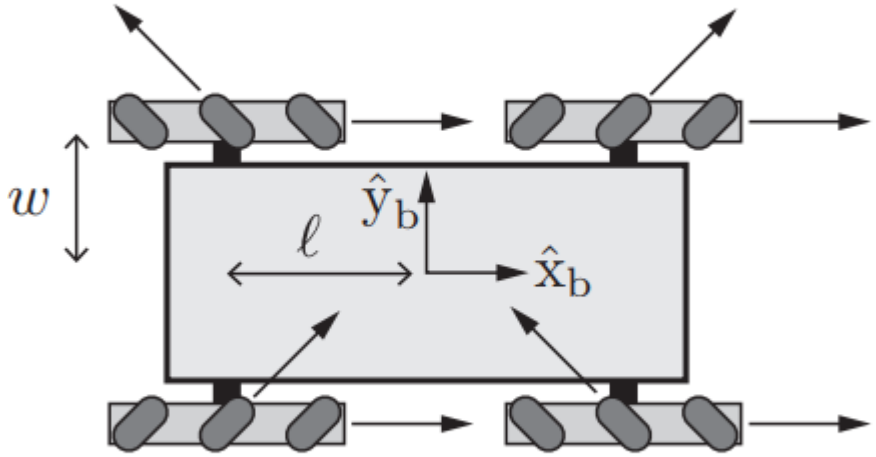
A triangular robot with Swedish wheels





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A robot with 4 Swedish wheels

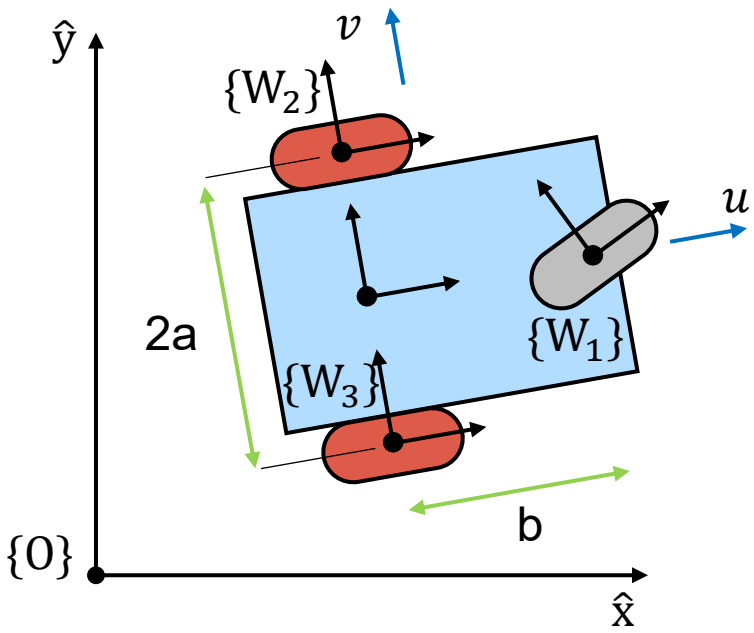


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Tricycle robot

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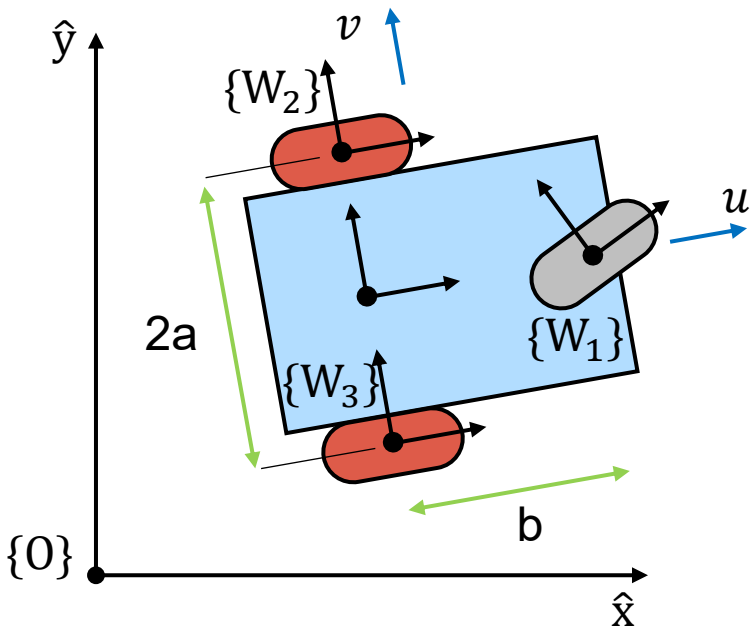
Tricycle robot



Standard wheels with no rollers.
Wheel radius is r.
Front wheel is driven and steered.
Rear wheels are not driven.

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Tricycle robot



$$r_1 = r$$
$$\gamma_1 = 0$$
$$\beta_1 = \beta_1(t)$$
$${}^{c}x_{W_1} = b$$
$${}^{c}y_{W_1} = 0$$

$$\dot{\varphi}_1 = \begin{bmatrix} \frac{1}{r} & 0 \end{bmatrix} \begin{bmatrix} \cos\beta_1 & \sin\beta_1 \\ -\sin\beta_1 & \cos\beta_1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & b \end{bmatrix} \begin{bmatrix} u \\ v \\ \dot{\theta} \end{bmatrix}$$

The rear wheels cannot slide laterally. So, v = 0

$${}^{c}v_{W_1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & b \end{bmatrix} \begin{bmatrix} u \\ 0 \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} u \\ b\dot{\theta} \end{bmatrix}$$ ← Recall step 2

$${}^{c}v_{W_1} = \begin{bmatrix} \cos\beta_1 & -\sin\beta_1 \\ \sin\beta_1 & \cos\beta_1 \end{bmatrix} \begin{bmatrix} r\dot{\varphi}_1 \\ 0 \end{bmatrix}$$ ← Recall step 1

Standard wheels with no rollers.
Wheel radius is r.
Front wheel is driven and steered.
Rear wheels are not driven.

This gives

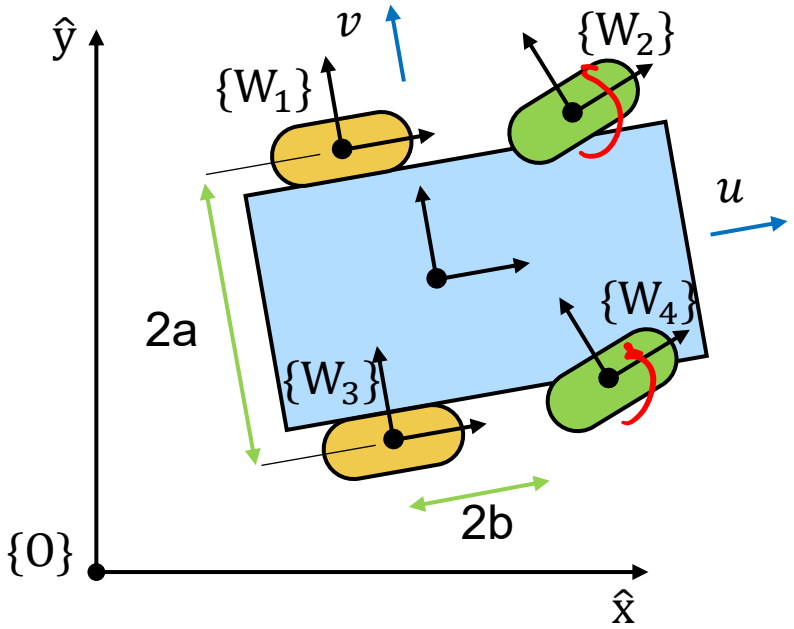
$$\dot{\varphi}_1 = \frac{1}{r}\sqrt{u^2 + b^2\dot{\theta}^2} \qquad \beta_1 = atan2(b\dot{\theta}, u)$$

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Homework: Car type robot

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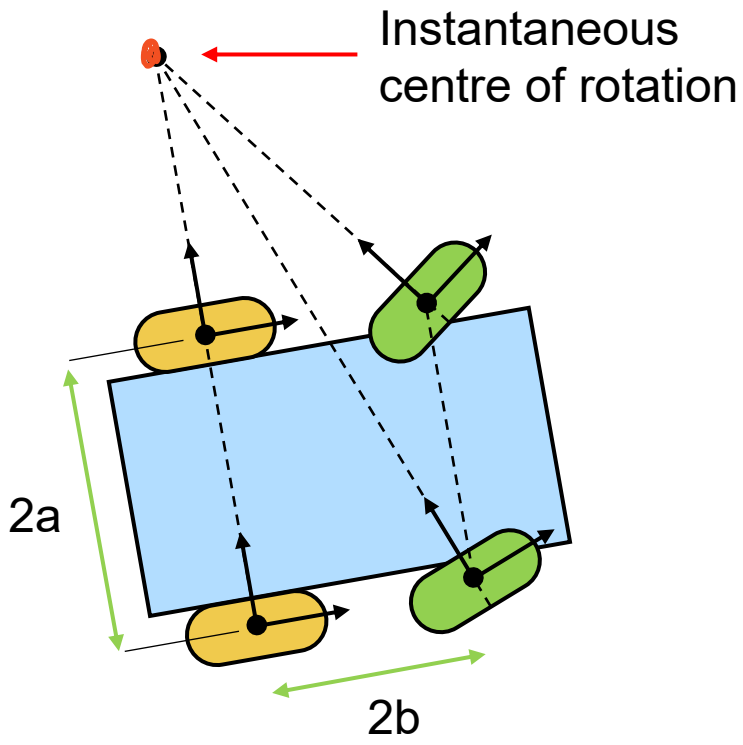
Homework: Car type robot



Standard wheels with no rollers.
Wheel radius is r.
Rear wheels are driven, cannot be steered.
Front wheels are steered, and not driven.

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Ackermann steering



The two front wheels are steered at different angles using an Ackermann steering. Its purpose is to make sure that all the wheels rotate without slipping.

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Maneuverability of a mobile robot

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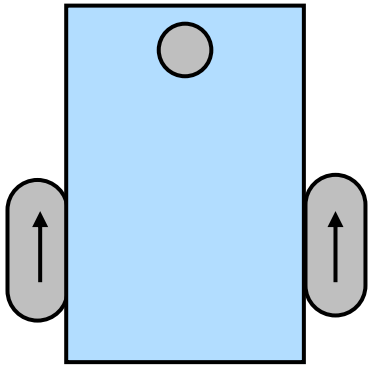
Maneuverability of a mobile robot

- The overall degrees of freedom that a robot can manipulate is called the degree of maneuverability, δ_{M}.
- This is defined in terms to two quantities:
  - mobility (not to be confused with the term we used in manipulator arms)
  - steerability
- Mobility – think of this as the number of driven wheels
- Steerability – think of this as the number of steerable wheels

$$\delta_{\mathrm{M}} = \delta_{\mathrm{m}} + \delta_{\mathrm{s}}$$

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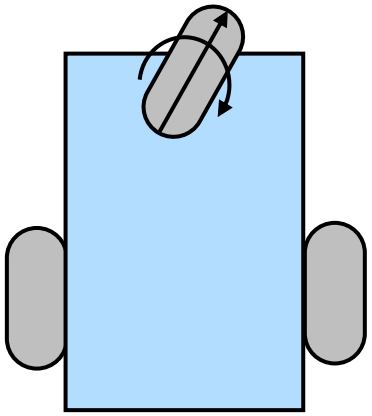
Examples



$\delta_m = 2$

$\delta_s = 0$

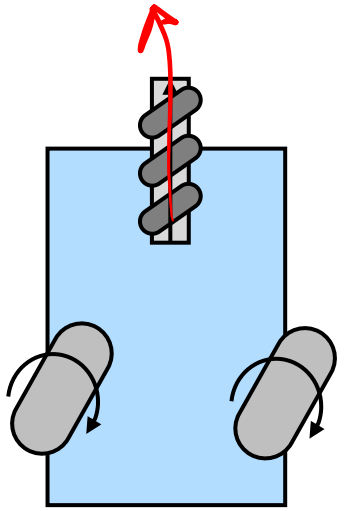
$\delta_M = 2$



$\delta_m = 1$

$\delta_s = 1$

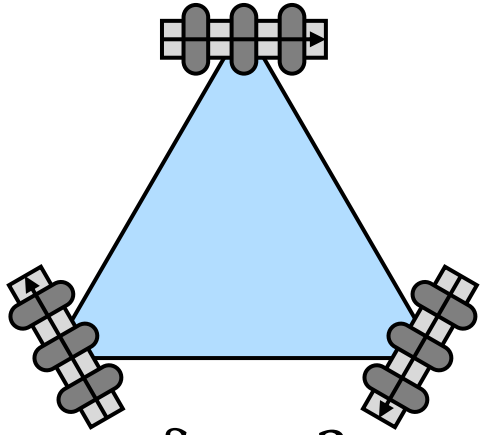
$\delta_M = 2$



$\delta_m = 1$

$\delta_s = 2$

$\delta_M = 3$



$\delta_m = 3$

$\delta_s = 0$

$\delta_M = 3$

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Constraints

Holonomic vs nonholonomic systems

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Holonomic kinematic constraint

$x^2 + y^2 = 1$

- A constraint that can be expressed explicitly as a function of position only is called holonomic.
- For example, a constraint of the type $f(\beta, \gamma, \varphi, \theta, \mathrm{x}, \mathrm{y}) = 0$ is holonomic.
- It must not contain derivatives of these quantities.

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Non-holonomic kinematic constraint

- These types of constraints contain derivatives of position terms.
- They cannot be integrated to a form that contains only position terms.
- They are also called non-integrable systems.

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Consider the kinematics of a standard wheel robot

$$\dot{\varphi}_i = \begin{bmatrix} \frac{1}{r_i} & 0 \end{bmatrix} \begin{bmatrix} \cos\beta_i & \sin\beta_i \\ -\sin\beta_i & \cos\beta_i \end{bmatrix} \begin{bmatrix} 1 & 0 & -{}^{c}y_{w_i} \\ 0 & 1 & {}^{c}x_{w_i} \end{bmatrix} \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix}$$

$$r_i\dot{\varphi}_i = \begin{bmatrix} \cos\beta_i & \sin\beta_i & -{}^{c}y_{w_i}\cos\beta_i + {}^{c}x_{w_i}\sin\beta_i \end{bmatrix} R^{-1}(\theta)\nu$$

$$0 = \begin{bmatrix} -\sin\beta_i & \cos\beta_i & {}^{c}y_{w_i}\sin\beta_i + {}^{c}x_{w_i}\cos\beta_i \end{bmatrix} R^{-1}(\theta)\nu$$

These are non-holonomic constraints

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Homework

- What about a bicycle? Rear wheel is driven, front wheel is steered.
- What about a bicycle whose steering is jammed in the central position?

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Summary

- We discussed the kinematics problem in the context of wheeled robots
- Generalised model of a wheel
- Different types of wheels and robots with wheels
- Constraints imposed by wheels
- Maneuverability of a mobile robot
- Holonomic and non-holonomic constraints

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Next lecture

- Dynamics of wheeled robots
- There is no class on 05.10.2026. The lecture will be recorded.

- Week 12 – Planning algorithms
- Week 13 – Algorithms continued and exam revision.

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